

Odoo Process Memory Gateway Architecture

The process memory gateway bridging AI agents with enterprise ERPs. Captures team conventions from conversation, enforces human review, and injects scoped policy directly into model prompts.



Dynamic Memory with Human Approval

Eliminating model amnesia and prompt stuffing without writing custom code.

1

Conversational Ingestion — Plain language statement in team chat

process_remember_instruction

2

Staging Quarantine — Non-enforceable; candidate isolated

status: pending_review

3

Human Approval Gate — Owner reviews, edits, or approves

process_review_candidate

4

Context-Scoped Injection — Injects matching policy into Bedrock

process_get_context

WITHOUT OPM (TRADITIONAL APPROACH)

- Hardcoded scripts:** Every policy check requires custom Python and redeploys.
- Model amnesia:** Models forget rules between sessions, violating standards.
- Prompt bloat:** 30-page SOPs dumped into every request trigger hallucinations.

WITH OPM PROCESS MEMORY

- Zero-code updates:** Leads define enforceable rules in plain English.
- Quarantine immunity:** 0 unapproved rules reach model prompts or ERP.
- Minimal token overhead:** Injects only relevant rules (~80 tokens vs 4,000+).

Key Outcome:

Business policies stay in company memory, not in custom code. Developers never modify gateway services for operational policy changes.

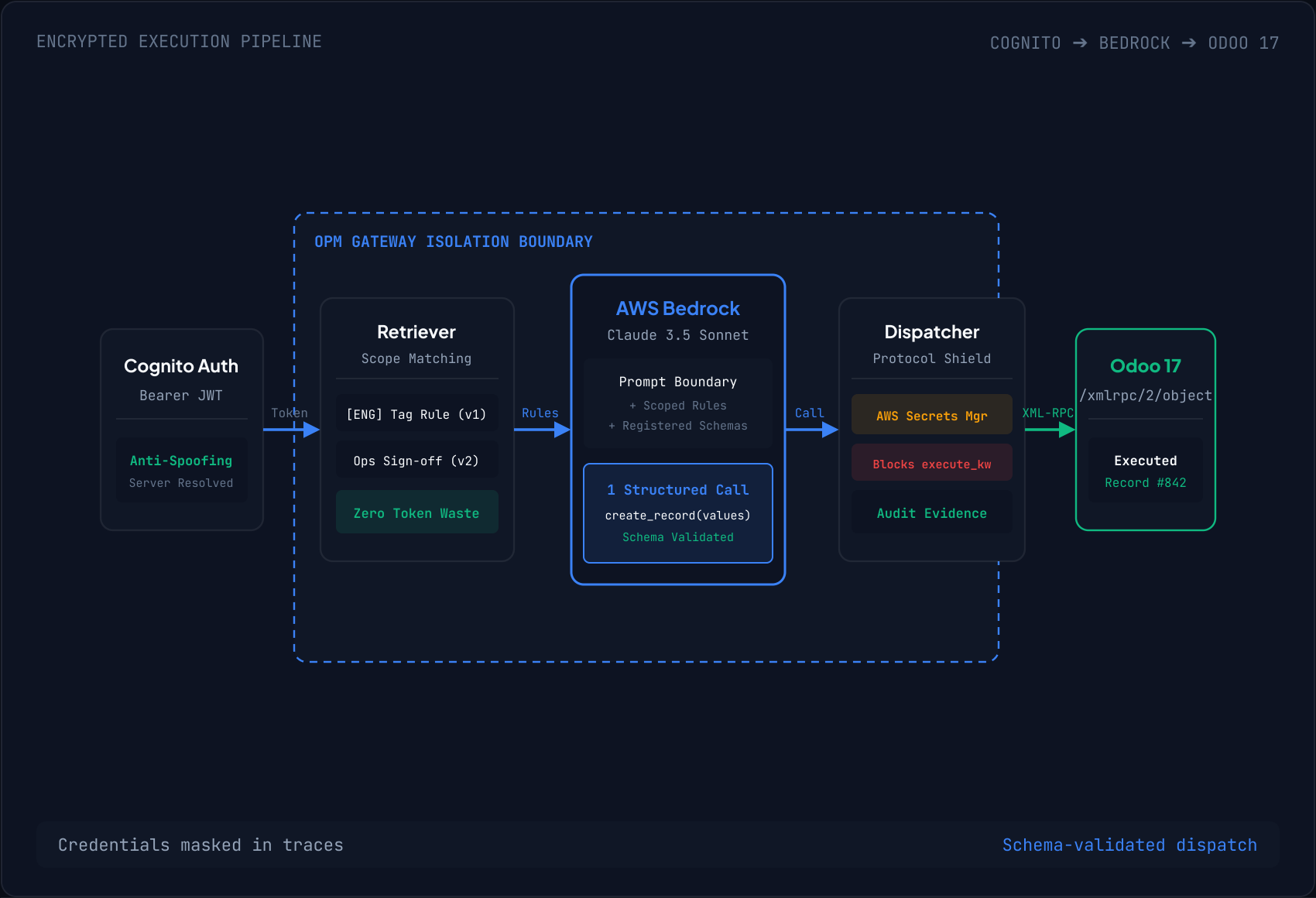
AUDITED CANONICAL RULE

CANONICAL v1 · APPROVED

```
{
  "rule_id": "rule_eng_prefix_001",
  "status": "approved",
  "rule_text": "Engineering tasks must start with [ENG] tag",
  "scope": { "system": "odoo", "resource": "project.task" },
  "reviewer": "lead@company.com"
}
```

Decoupled Gateway & Zero-Trust Dispatch

Protecting core ERP databases from unconstrained AI access without hardcoding business workflows.



1. Pure Gateway Decoupling

OPM contains zero hardcoded Odoo task recipes, Gantt formulas, or status workflows. It acts purely as a protocol and policy router.

- Standard FastMCP tool contracts
- Decoupled from ERP API shifts

2. Cryptographic Multi-Tenancy

Tenant and user identity is resolved cryptographically from Cognito JWTs. Client applications cannot spoof company IDs or roles.

- Zero client-controlled identity
- Owner/reviewer role enforcement

3. Credential Masking & Audit Trace

Downstream ERP passwords are resolved via AWS Secrets Manager and masked from logs. Unsafe raw endpoints like execute_kw are blocked.

- Model never sees credentials
- 100% deterministic trace telemetry